

Randall West

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Principal Software Engineer

Principal Software Engineer with 20+ years of experience building and operating high-scale distributed systems in Linux environments, with a strong focus on real-time data processing, system reliability, automation, and production debugging. Proven track record delivering software that powers large-scale deployments and supports millions of users in production.

Experienced in developing backend services, system integration, automation tooling, and network management solutions for large-scale telecommunications platforms. Strong background in C/C++, Python, and Go development, with hands-on experience supporting VMware infrastructure through PowerCLI automation for vCenter and ESXi environments.

Actively pursuing full-stack software development through the design and implementation of Ref Ledger, a web-based officiating management application featuring a browser-based frontend, Go backend services, and MongoDB database integration. Expanding expertise in modern cloud-native technologies including Docker and Kubernetes.

- Strong background in distributed systems, real-time data flows, and event-driven architectures
- Deep experience with system reliability, debugging, and production issue resolution at scale
- Proficient in C/C++, Python, and Go, with active development of backend services and web applications
- Experience building APIs, CLI tools, automation frameworks, and control systems for large-scale deployments
- Developed PowerCLI automation solutions for monitoring and maintaining VMware vCenter and ESXi server environments
- Designing and developing a full-stack web application with frontend GUI, backend services, and MongoDB database integration
- Skilled at translating real-world operational needs into scalable software solutions
- Proficient in Windows OS
- Proficient in Jira and Confluence

- Collaborative across engineering, QA, operations, and customer-facing teams

Technical Skills

Languages: C, C++, Python, Go (actively developing backend services)

Systems: Linux, VMware vCenter, ESXi, Windows OS

Distributed Systems: High-scale system design, multithreading, performance optimization

Networking: TCP/UDP, DNS, NAT, multicast, RADIUS, TACACS+, SSH

Web Technologies: HTML, CSS, JavaScript

APIs & Data: JSON, YAML

Issue Tracking Tools: GitHub, Jira, ClearQuest

Collaboration Platform: Confluence

Containers & Orchestration: Docker, Kubernetes (in progress)

Automation: Python, Perl, TCL, PowerShell

Database: MySQL, MongoDB

Core Strengths: Real-time systems, backend services, system integration, production debugging, scalability

Professional Development

- Completed hands-on training in Docker, building containerized services and deployment workflows
- Actively developing Kubernetes-based deployment knowledge (service orchestration, scaling, resilience)
- Designing and developing Ref Ledger, a full-stack officiating management application with a web-based frontend, Go backend services, and MongoDB database
- Building proficiency in Go for backend service development and distributed systems tooling
- Applying containerization and service-based architecture concepts to personal and professional projects
- Completed course on MongoDB for SQL Experts
- Studying for CKA certification

Work Experience

COMMSCOPE — Principal Software Engineer

Suwanee, GA / 2019 – 2026

- Developed and maintained C/C++ command-line interface (CLI) functionality for multiple cable access platforms, including the E6000, C4, and RMD, enabling configuration, management, and troubleshooting of system features.
- Designed and developed a cable modem simulation application in C/C++ used for scalability and performance testing of the E6000, C4, and RMD platforms, supporting validation of system capacity, reliability, and large-scale deployment scenarios.
- Developed PowerCLI automation scripts to monitor, maintain, and manage VMware infrastructure, including vCenter and ESXi server environments, improving operational efficiency and system visibility.
- Designed and developed software for large-scale distributed systems supporting 10M+ users across 100+ deployments
- Built real-time control and configuration systems used to manage and monitor distributed network infrastructure
- Developed backend logic and system components responsible for processing and reacting to high-volume operational events
- Created Python-based automation frameworks to improve deployment reliability, validation, and system observability
- Implemented multithreaded and performance-optimized components to meet real-time system demands
- Led root cause analysis of complex production issues, improving system stability and reducing operational downtime
- Collaborated cross-functionally with engineering, QA, and operations to deliver features used in live production environments
- Supported deployment and integration efforts, ensuring successful rollout of systems in customer environments

ARRIS GROUP — Senior Software Engineer

Suwanee, GA | 1997 – 2019

- Developed and maintained C/C++ command-line interface (CLI) functionality for multiple cable access platforms, including the E6000, C4, and RMD, enabling configuration, management, and troubleshooting of system features.
- Designed and developed a cable modem simulation application in C/C++ used for scalability and performance testing of the E6000, C4, and RMD platforms, supporting validation of system capacity, reliability, and large-scale deployment scenarios.
- Developed C/C++ diagnostic CLI commands for the C4 platform to validate and troubleshoot hardware components, enabling testing and verification of critical system chips during development and production support.
- Designed and implemented hardware diagnostic utilities that exercised onboard chipsets and subsystems, improving fault isolation, troubleshooting efficiency, and system reliability.

- Developed high-performance software for distributed networking and broadband systems
- Built control plane interfaces and backend logic for provisioning, diagnostics, and system management
- Delivered integration solutions across multiple system components, ensuring reliable end-to-end system behavior
- Diagnosed and resolved complex system issues through deep debugging, log analysis, and code tracing
- Optimized system performance, improving throughput, latency, and scalability under load
- Refactored legacy systems to support long-term maintainability and evolving system requirements
- Contributed to deployment support and customer-facing issue resolution in production environments

Education

Bachelor of Science, Systems Science (Dean's List)

University of West Florida

Key Achievements

- Designed and implemented a scalable testing platform, improving system validation and reducing operational cost
- Co-inventor, U.S. Patent #7,574,345
- Multiple engineering excellence awards (Arris)